

DEXTER FERNANDES

Data Scientist | Computer Vision & Machine Learning Bristol, UK | Open to relocate to Manchester +44 7436 007369

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SUMMARY

Data Scientist and Computer Vision Engineer with 3+ years delivering machine learning models from data preparation and experimentation through cloud and on-premise edge deployment. Strong Python, PyTorch, TensorFlow, scikit-learn and statistical model validation across object detection and multivariate time-series classification. Owned 24/7 industrial AI systems for international clients and led cross-functional delivery. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar - AI Engineer

01/2023 - 08/2023 | MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for 24/7 computer vision on BP drilling rigs, from data curation and PyTorch training through deployment and on-site validation, improving F1 by 15%.
- Built image, video and sensor ingestion and validation pipelines with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 72 hours to under 24 hours.
- Adapted YOLOv5 neural networks in PyTorch for rig-specific classes and ran Optuna sweeps with Weights & Biases experiment tracking to support algorithm selection and optimisation.
- Benchmarked YOLOv5 against EfficientDet on a repeatable harness, measuring accuracy, latency and stability before and after optimisation to catch regressions.
- Applied OpenCV and scikit-image preprocessing and artefact detection to contaminated camera feeds, reducing false positives by 13%.
- Developed multivariate time-series classifiers over 91 sensor channels using MultiRocket, gradient boosting and random forests.
- Applied Bayesian optimisation, ablation studies, hypothesis testing and confidence intervals in Jupyter for model validation, threshold tuning and failure analysis.
- Hardened DeepStream services with Dockerised microservices, Kafka, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Owned post-deployment monitoring and drift tracking for fielded vision systems, feeding production failure analysis back into retraining.
- Presented model and deployment findings to leadership, project managers and rig operators, translating production data into rollout and optimisation decisions.

LivNSense Technologies - Lead Software Engineer, AI

05/2020 - 12/2022 | BENGALURU, INDIA

- Led four engineers delivering a two-stage detector and helmet-colour classifier for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Owned production ML pipelines from ingestion and labelling through training, optimisation, deployment, monitoring and field validation for industrial safety systems.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and industrial Core i7 edge systems, increasing throughput from 5 to 15 FPS.
- Optimised neural network architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle and added ByteTrack to maintain worker and vehicle identities across frames.
- Ran training and validation under occlusion, lighting variation and motion artefacts, reducing performance variance by 25% across camera conditions.
- Automated labelling with existing detection models and DVC dataset versioning, reducing manual labelling effort by 80% on large real-world datasets.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based asynchronous processing, owning autoscaling, monitoring and cost-per-inference.
- Evaluated edge against cloud inference to balance bandwidth, latency and operational cost while maintaining safety-critical performance targets.
- Designed a PostgreSQL schema for real-time incident analytics and wrote the joins and CTEs used in operational reporting across cross-functional teams.

PROJECTS

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Developed and evaluated a repeatable Python benchmarking harness for PyTorch detectors including YOLOv8, YOLOv10, YOLOv26 and RT-DETR.
- Compared accuracy, latency and model size, then exported models to ONNX and tested OpenVINO deployment paths for low-latency targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2, assessing segmentation and depth models under the same edge constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.

DeepStream RTSP Pipeline

github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built a DeepStream 9.0 and GStreamer pipeline for YOLO detection and NvDCF tracking across three concurrent RTSP feeds, with metadata extraction and anonymisation.
- Converted YOLO26n to a TensorRT FP16 engine supporting variable batch sizes, validated FP16 against FP32 accuracy and profiled p99 latency to sustain 29.7 FPS per stream live.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM impact to guide model selection.

ROS2 SLAM and Nav2 Simulation

02/2026

- Developed and evaluated a simulation-based robotics stack in Python, C++, ROS2 Jazzy and Gazebo Harmonic for LiDAR SLAM and autonomous navigation.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations to support algorithm selection.

EDUCATION

University of Bristol - MSc Robotics, Merit

09/2023 - 09/2024 | BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built and evaluated a real-time C++17 and ROS2 SLAM system on a Clearpath Husky, fusing LiDAR, depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer on matched rosbag2 sequences under a common evaluation protocol.
- Configured Ceres, g2o and GTSAM solver backends and tuned ICP parameters across environments.

Visvesvaraya National Institute of Technology - BTech Electrical and Electronics Engineering

07/2016 - 05/2020 | NAGPUR, INDIA

- Studied signals, systems, embedded hardware and control, forming a foundation for real-time AI, sensor systems and robotics software.

SKILLS

Programming and data	Python, SQL, C++17, Bash, NumPy, Pandas, scikit-learn, Jupyter, PostgreSQL
ML and statistics	Classification, neural networks, gradient boosting, random forests, MultiRocket, hypothesis testing, confidence intervals, Bayesian optimisation, ablation studies
Computer vision	Object detection, image classification, segmentation, tracking, YOLO, RT-DETR, EfficientDet, OpenCV, scikit-image
Deep learning	PyTorch, TensorFlow, Keras, convolutional neural networks (CNNs), data augmentation, synthetic data
Data and experiments	Data preparation, preprocessing, curation, model validation, failure analysis, benchmarking, CVAT, TIDE
MLOps	Docker, MLflow, Apache Airflow, DVC, Optuna, Weights & Biases, Git, CI/CD, pytest
Deployment and cloud	Microsoft Azure, on-premise edge deployment, REST APIs, containerised microservices, Kafka, production monitoring
Edge AI	NVIDIA DeepStream, GStreamer, TensorRT, OpenVINO, ONNX, FP16/INT8, quantisation, pruning
Professional	Technical leadership, mentoring, stakeholder communication, cross-functional delivery, Agile, Scrum

ADDITIONAL EXPERIENCE

GrowthStage - Technical Assessor, Computer Vision and Software Engineering

05/2026 - PRESENT | REMOTE

Assess candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)